Module I: Introduction

1.1 Professional software development

Lots of people write programs. People in business write spreadsheet programs to simplify their jobs, scientists and engineers write programs to process their experimental data, and hobbyists write programs for their own interest and enjoyment. However, the vast majority of software development is a professional activity where software is developed for specific business purposes, for inclusion in other devices, or as software products such as information systems, CAD systems, etc. Professional software, intended for use by someone apart from its developer, is usually developed by teams rather than individuals. It is maintained and changed throughout its life.

Software engineering is intended to support professional software development, rather than individual programming. It includes techniques that support program specification, design, and evolution, none of which are normally relevant for personal software development. Frequently asked questions about software engineering is summarized in Figure 1.1.

Many people think that software is simply another word for computer programs. However, when we are talking about software engineering, software is not just the programs themselves but also all associated documentation and configuration data that is required to make these programs operate correctly. A professionally developed software system is often more than a single program. The system usually consists of a number of separate programs and configuration files that are used to set up these programs. It may include system documentation, which describes the structure of the system; user documentation, which explains how to use the system, and websites for users to download recent product information.

Software engineers are concerned with developing software products (i.e., software which can be sold to a customer). There are two kinds of software products:

1. **Generic products** These are stand-alone systems that are produced by a development organization and sold on the open market to any customer who is able to buy them. **Examples** of this type of product include software for PCs such as databases, word processors, drawing packages, and project-management tools. It also includes so-called vertical applications designed for some specific purpose such as library information systems, accounting systems, or systems for maintaining dental records.
2. **Customized (or bespoke) products** These are systems that are commissioned by a particular customer. A software contractor develops the software especially for that customer. **Examples** of this type of software include control systems for electronic devices, systems written to support a particular business process, and air traffic control systems.

Question	Answer
What is software?	Computer programs and associated documentation. Software products may be developed for a particular customer or may be developed for a general market.
What are the attributes of good software?	Good software should deliver the required functionality and performance to the user and should be maintainable, dependable, and usable.
What is software engineering?	Software engineering is an engineering discipline that is concerned with all aspects of software production.
What are the fundamental software engineering activities?	Software specification, software development, software validation, and software evolution.
What is the difference between software engineering and computer science?	Computer science focuses on theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.
What is the difference between software engineering and system engineering?	System engineering is concerned with all aspects of computer-based systems development including hardware, software, and process engineering. Software engineering is part of this more general process.
What are the key challenges facing software engineering?	Coping with increasing diversity, demands for reduced delivery times, and developing trustworthy software.
What are the costs of software engineering?	Roughly 60% of software costs are development costs; 40% are testing costs. For custom software, evolution costs often exceed development costs.
What are the best software engineering techniques and methods?	While all software projects have to be professionally managed and developed, different techniques are appropriate for different types of system. For example, games should always be developed using a series of prototypes whereas safety critical control systems require a complete and analyzable specification to be developed. You can't, therefore, say that one method is better than another.
What differences has the Web made to software engineering?	The Web has led to the availability of software services and the possibility of developing highly distributed service-based systems. Web-based systems development has led to important advances in programming languages and software reuse.

Fig.1.1: Frequently asked questions about software

An important difference between these types of software is that, in generic products, the organization that develops the software controls the software specification. For custom products, the specification is usually developed and controlled by the organization that is buying the software. The software developers must work to that specification.

The quality of professional software

Quality is therefore not just concerned with what the software does. Rather, it has to include the software's behavior while it is executing and the structure and organization of the system programs and associated documentation. This is reflected in so-called **quality or non-functional software attributes**.

Examples of these attributes are the software's **response time** to a user query and the **understandability** of the program code. The specific set of attributes that you might expect from a software system obviously depends on its application. Therefore, a banking system must be secure, an interactive game must be responsive, a telephone switching system must be reliable, and so on. These can be generalized into the set of attributes shown in Figure 1.2.

Product characteristics	Description
Maintainability	Software should be written in such a way so that it can evolve to meet the changing needs of customers. This is a critical attribute because software change is an inevitable requirement of a changing business environment.
Dependability and security	Software dependability includes a range of characteristics including reliability, security, and safety. Dependable software should not cause physical or economic damage in the event of system failure. Malicious users should not be able to access or damage the system.
Efficiency	Software should not make wasteful use of system resources such as memory and processor cycles. Efficiency therefore includes responsiveness, processing time, memory utilization, etc.
Acceptability	Software must be acceptable to the type of users for which it is designed. This means that it must be understandable, usable, and compatible with other systems that they use.

Fig. 1.2: Essential attributes of good software

1.1.1 Software Engineering

Software engineering is an engineering discipline that is concerned with all aspects of software production from the early stages of system specification through to maintaining the system after it has gone into use. In this definition, there are **two key phrases**:

1. **Engineering discipline** Engineers make things work. They apply theories, methods, and tools where these are appropriate. However, they use them selectively and always try to discover solutions to problems even when there are no applicable theories and methods. Engineers also recognize that they must work to organizational and financial constraints so they look for solutions within these constraints.
2. **All aspects of software production** Software engineering is not just concerned with the technical processes of software development. It also includes activities such as software project management and the development of tools, methods, and theories to support software production.

Why Software engineering is important?

1. More and more, individuals and society rely on advanced software systems. We need to be able to produce reliable and trustworthy systems economically and quickly.
2. It is usually cheaper, in the long run, to use software engineering methods and techniques for software systems rather than just write the programs as if it was a personal programming project. For most types of systems, the majority of costs are the costs of changing the software after it has gone into use.

Software process

The systematic approach that is used in software engineering is sometimes called a **software process.** A software process is a sequence of activities that leads to the production of a software product. There are four fundamental activities that are common to all software processes. These activities are:

1. **Software specification**, where customers and engineers define the software that is to be produced and the constraints on its operation.
2. **Software development**, where the software is designed and programmed.
3. **Software validation**, where the software is checked to ensure that it is what the customer requires.
4. **Software evolution**, where the software is modified to reflect changing customer and market requirements.

How software engineering is related to both computer science and systems engineering?

Computer science is concerned with the theories and methods that underlie computers and software systems, whereas software engineering is concerned with the practical problems of producing software. Some knowledge of computer science is essential for software engineers in the same way that some knowledge of physics is essential for electrical engineers. Computer science theory, however, is often most applicable to relatively small programs. Elegant theories of computer science cannot always be applied to large, complex problems that require a software solution.

System engineering is concerned with all aspects of the development and evolution of complex systems where software plays a major role. System engineering is therefore concerned with hardware development, policy and process design and system deployment, as well as software engineering. System engineers are involved in specifying the system, defining its overall architecture, and then integrating the different parts to create the finished system. They are less concerned with the engineering of the system components (hardware, software, etc.).

Which are the general issues that affect many different types of software?

1. **Heterogeneity** Increasingly, systems are required to operate as distributed systems across networks that include different types of computer and mobile devices. As well

as running on general-purpose computers, software may also have to execute on mobile phones. You often have to integrate new software with older legacy systems written in different programming languages. The challenge here is to develop techniques for building dependable software that is flexible enough to cope with this heterogeneity.

2. **Business and social change** Business and society are changing incredibly quickly as emerging economies develop and new technologies become available. They need to be able to change their existing software and to rapidly develop new software. Many traditional software engineering techniques are time consuming and delivery of new systems often takes longer than planned. They need to evolve so that the time required for software to deliver value to its customers is reduced.
3. **Security and trust as software** is intertwined with all aspects of our lives, it is essential that we can trust that software. This is especially true for remote software systems accessed through a web page or web service interface. We have to make sure that malicious users cannot attack our software and that information security is maintained.

1.1.2 Software engineering diversity

Perhaps the most significant factor in determining which software engineering methods and techniques are most important is the type of application that is being developed. There are many **different types of application** including:

1. **Stand-alone applications** These are application systems that run on a local computer, such as a PC. They include all necessary functionality and do not need to be connected to a network. Examples of such applications are office applications on a PC, CAD programs, photo manipulation software, etc.
2. **Interactive transaction-based applications** These are applications that execute on a remote computer and that are accessed by users from their own PCs or terminals. Obviously, these include web applications such as e-commerce applications where you can interact with a remote system to buy goods and services. This class of application also includes business systems, where a business provides access to its systems through a web browser or special-purpose client program and cloud-based services, such as mail and photo sharing. Interactive applications often incorporate a large data store that is accessed and updated in each transaction.
3. **Embedded control systems** These are software control systems that control and manage hardware devices. Numerically, there are probably more embedded systems than any other type of system. Examples of embedded systems include the software in a mobile (cell) phone, software that controls anti-lock braking in a car, and software in a microwave oven to control the cooking process.
4. **Batch processing systems** These are business systems that are designed to process data in large batches. They process large numbers of individual inputs to create

corresponding outputs. Examples of batch systems include periodic billing systems, such as phone billing systems, and salary payment systems.

5. **Entertainment systems** These are systems that are primarily for personal use and which are intended to entertain the user. Most of these systems are games of one kind or another. The quality of the user interaction offered is the most important distinguishing characteristic of entertainment systems.
6. **Systems for modeling and simulation** These are systems that are developed by scientists and engineers to model physical processes or situations, which include many, separate, interacting objects. These are often computationally intensive and require high-performance parallel systems for execution.
7. **Data collection systems** These are systems that collect data from their environment using a set of sensors and send that data to other systems for processing. The software has to interact with sensors and often is installed in a hostile environment such as inside an engine or in a remote location.
8. **Systems of systems** These are systems that are composed of a number of other software systems. Some of these may be generic software products, such as a spreadsheet program. Other systems in the assembly may be specially written for that environment.

Which are the software engineering fundamentals that apply to all types of software system?

1. They should be developed using a managed and **understood development process**. The organization developing the software should plan the development process and have clear ideas of what will be produced and when it will be completed. Of course, different processes are used for different types of software.
2. **Dependability and performance** are important for all types of systems. Software should behave as expected, without failures and should be available for use when it is required. It should be safe in its operation and, as far as possible, should be secure against external attack. The system should perform efficiently and should not waste resources.
3. **Understanding and managing the software specification and requirements** (what the software should do) are important. You have to know what different customers and users of the system expect from it and you have to manage their expectations so that a useful system can be delivered within budget and to schedule.
4. You should make as effective use as possible of existing resources. This means that, where appropriate, you should **reuse software** that has already been developed rather than write new software.

1.1.3 Software engineering and the Web

Initially, the Web was primarily a universally accessible information store and it had little effect on software systems. These systems ran on local computers and were only accessible from within an organization. Around 2000, the Web started to evolve and more and more

functionality was added to browsers. This meant that web-based systems could be developed where, instead of a special-purpose user interface, these systems could be accessed using a web browser. This led to the development of a vast range of new system products that delivered innovative services, accessed over the Web.

As well as these system products, the development of web browsers that could run small programs and do some local processing led to an evolution in business and organizational software. Instead of writing software and deploying it on users' PCs, the software was deployed on a web server. This made it much cheaper to change and upgrade the software, as there was no need to install the software on every PC. It also reduced costs, as user interface development is particularly expensive.

The next stage in the development of web-based systems was the notion of web services. Web services are software components that deliver specific, useful functionality and which are accessed over the Web. Applications are constructed by integrating these web services, which may be provided by different companies.

In the last few years, the notion of 'software as a service' has been developed. It has been proposed that software will not normally run on local computers but will run on 'computing clouds' that are accessed over the Internet. If you use a service such as web-based mail, you are using a cloud-based system. A **computing cloud** is a huge number of linked computer systems that is shared by many users. Users do not buy software but pay according to how much the software is used or are given free access in return for watching adverts that are displayed on their screen.

How web-based systems are engineered?

1. Software reuse has become the dominant approach for constructing web-based systems. When building these systems, you think about how you can assemble them from pre-existing software components and systems.
2. It is now generally recognized that it is impractical to specify all the requirements for such systems in advance. Web-based systems should be developed and delivered incrementally.
3. User interfaces are constrained by the capabilities of web browsers. Although technologies such as AJAX mean that rich interfaces can be created within a web browser, these technologies are still difficult to use.

1.2 Software engineering ethics

Professional responsibilities

1. **Confidentiality** You should normally respect the confidentiality of your employers or clients irrespective of whether or not a formal confidentiality agreement has been signed.
2. **Competence** You should not misrepresent your level of competence. You should not knowingly accept work that is outside your competence.

3. **Intellectual property rights** You should be aware of local laws governing the use of intellectual property such as patents and copyright. You should be careful to ensure that the intellectual property of employers and clients is protected.
4. **Computer misuse** You should not use your technical skills to misuse other people's computers. Computer misuse ranges from relatively trivial (game playing on an employer's machine, say) to extremely serious (dissemination of viruses or other malware).

1.3 Case studies

An insulin pump control system

An insulin pump is a medical system that simulates the operation of the pancreas (an internal organ). The software controlling this system is an embedded system, which collects information from a sensor and controls a pump that delivers a controlled dose of insulin to a user.

People who suffer from diabetes use the system. Diabetes is a relatively common condition where the human pancreas is unable to produce sufficient quantities of a hormone called insulin. Insulin metabolizes glucose (sugar) in the blood. The conventional treatment of diabetes involves regular injections of genetically engineered insulin. Diabetics measure their blood sugar levels using an external meter and then calculate the dose of insulin that they should inject.

The problem with this treatment is that the level of insulin required does not just depend on the blood glucose level but also on the time of the last insulin injection. This can lead to very low levels of blood glucose (if there is too much insulin) or very high levels of blood sugar (if there is too little insulin). Low blood glucose is, in the short term, a more serious condition as it can result in temporary brain malfunctioning and, ultimately, unconsciousness and death. In the long term, however, continual high levels of blood glucose can lead to eye damage, kidney damage, and heart problems.

Current advances in developing miniaturized sensors have meant that it is now possible to develop automated insulin delivery systems. These systems monitor blood sugar levels and deliver an appropriate dose of insulin when required. Insulin delivery systems like this already exist for the treatment of hospital patients. In the future, it may be possible for many diabetics to have such systems permanently attached to their bodies.

A software-controlled insulin delivery system might work by using a micro sensor embedded in the patient to measure some blood parameter that is proportional to the sugar level. This is then sent to the pump controller. This controller computes the sugar level and the amount of insulin that is needed. It then sends signals to a miniaturized pump to deliver the insulin via a permanently attached needle.

Figure 1.3 shows the hardware components and organization of the insulin pump. Figure 1.4 is a UML activity model that illustrates how the software transforms an input blood sugar level to a sequence of commands that drive the insulin pump.

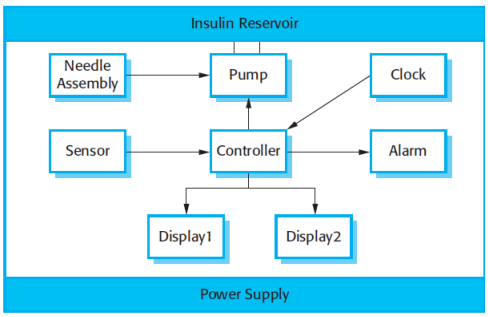


Fig. 1.3: Insulin pump hardware

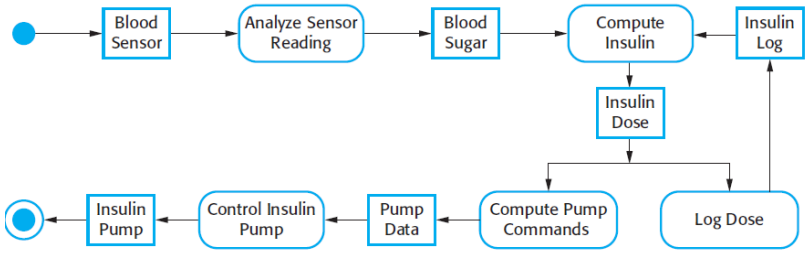


Fig. 1.4: Activity model of the insulin pump

Clearly, this is a safety-critical system. If the pump fails to operate or does not operate correctly, then the user's health may be damaged or they may fall into a coma because their blood sugar levels are too high or too low. There are, therefore, two essential high-level requirements that this system must meet:

1. The system shall be available to deliver insulin when required.

2. The system shall perform reliably and deliver the correct amount of insulin to counteract the current level of blood sugar.

A patient information system for mental health care

A patient information system to support mental health care is a medical information system that maintains information about patients suffering from mental health problems and the treatments that they have received. Most mental health patients do not require dedicated hospital treatment but need to attend specialist clinics regularly where they can meet a doctor who has detailed knowledge of their problems. To make it easier for patients to attend, these clinics are not just run in hospitals. They may also be held in local medical practices or community centers.

The **MHC-PMS** (Mental Health Care-Patient Management System) is an information system that is intended for use in clinics. It makes use of a centralized database of patient information but has also been designed to run on a PC, so that it may be accessed and used from sites that do not have secure network connectivity. When the local systems have secure

network access, they use patient information in the database but they can download and use local copies of patient records when they are disconnected. The system is not a complete medical records system so does not maintain information about other medical conditions. However, it may interact and exchange data with other clinical information systems. Figure 1.5 illustrates the organization of the MHC-PMS.

The MHC-PMS has **two overall goals**:

1. To generate management information that allows health service managers to assess performance against local and government targets.
2. To provide medical staff with timely information to support the treatment of patients.

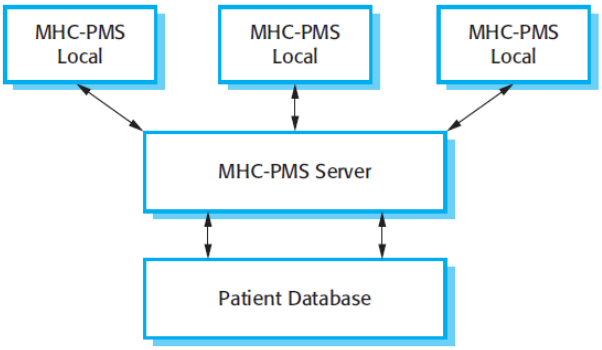


Fig 1.5: The organization of the MHC-PMS

A wilderness weather station

To help monitor climate change and to improve the accuracy of weather forecasts in remote areas, the government of a country with large areas of wilderness decides to deploy several hundred weather stations in remote areas. These weather stations collect data from a set of instruments that measure temperature and pressure, sunshine, rainfall, wind speed, and wind direction.

Wilderness weather stations are part of a larger system (Figure 1.6), which is a weather information system that collects data from weather stations and makes it available to other systems for processing. The systems in Figure 1.6 are:

1. The weather station system This is responsible for collecting weather data, carrying out some initial data processing, and transmitting it to the data management system.
2. The data management and archiving system This system collects the data from all of the wilderness weather stations, carries out data processing and analysis, and archives the data in a form that can be retrieved by other systems, such as weather forecasting systems.
3. The station maintenance system This system can communicate by satellite with all wilderness weather stations to monitor the health of these systems and provide reports of problems. It can update the embedded software in these systems. In the

event of system problems, this system can also be used to remotely control a wilderness weather system.

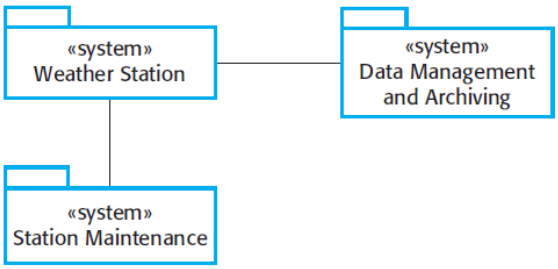


Fig. 1.6: The weather station's environment

In Figure 1.6, I have used the UML package symbol to indicate that each system is a collection of components and have identified the separate systems, using the UML stereotype «system». The associations between the packages indicate there is an exchange of information but, at this stage, there is no need to define them in any more detail.

1.4 The software process

A **software process** is a set of related activities that leads to the production of a software product. These activities may involve the development of software from scratch in a standard programming language like Java or C.

There are many different software processes but all must include four activities that are fundamental to software engineering:

1. Software specification The functionality of the software and constraints on its operation must be defined.
2. Software design and implementation The software to meet the specification must be produced.
3. Software validation The software must be validated to ensure that it does what the customer wants.
4. Software evolution The software must evolve to meet changing customer needs.

1.4.1 Software process models

The Waterfall Model

Because of the cascade from one phase to another, this model is known as the **'waterfall model'** or software life cycle. The waterfall model is an example of a plan- driven process. The principal stages of the waterfall model directly reflect the fundamental development activities:

1. **Requirements analysis and definition** The system's services, constraints, and goals are established by consultation with system users. They are then defined in detail and serve as a system specification.
2. **System and software design** The systems design process allocates the requirements to either hardware or software systems by establishing an overall system architecture. Software design involves identifying and describing the fundamental software system abstractions and their relationships.
3. **Implementation and unit testing** During this stage, the software design is realized as a set of programs or program units. Unit testing involves verifying that each unit meets its specification.
4. **Integration and system testing** The individual program units or programs are integrated and tested as a complete system to ensure that the software requirements have been met. After testing, the software system is delivered to the customer.
5. **Operation and maintenance** Normally, this is the longest life cycle phase. The system is installed and put into practical use. Maintenance involves correcting errors which were not discovered in earlier stages of the life cycle, improving the implementation of system units and enhancing the system's services as new requirements are discovered.

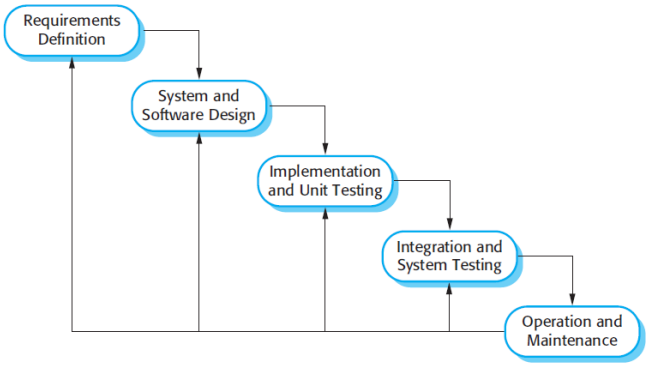


Fig. 1.7: The waterfall model

Waterfall model problems

1. Inflexible partitioning of the project into distinct stages makes it difficult to respond to changing customer requirements. Therefore, this model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design process.
2. The waterfall model is mostly used for large systems engineering projects where a system is developed at several sites.

Incremental Development

Incremental development is based on the idea of developing an initial implementation, exposing this to user comment and evolving it through several versions until an adequate system has been developed (Figure 1.8). Specification, development, and validation

activities are interleaved rather than separate, with rapid feedback across activities. Incremental software development, which is a fundamental part of agile approaches, is better than a waterfall approach for most business, e-commerce, and personal systems. Incremental development reflects the way that we solve problems.

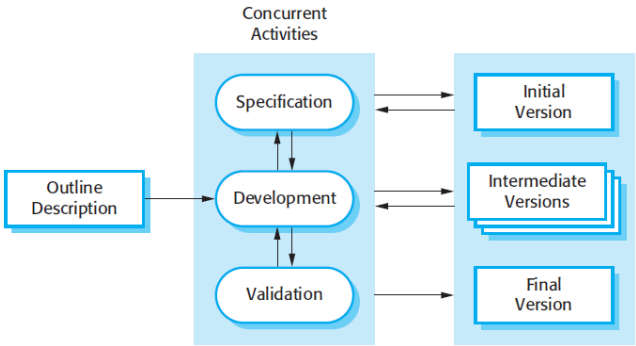


Fig. 1.8: Incremental development

Incremental development has **three important benefits**, compared to the waterfall model:

1. The cost of accommodating changing customer requirements is reduced. The amount of analysis and documentation that has to be redone is much less than is required with the waterfall model.
2. It is easier to get customer feedback on the development work that has been done. Customers can comment on demonstrations of the software and see how much has been implemented. Customers find it difficult to judge progress from software design documents.
3. More rapid delivery and deployment of useful software to the customer is possible, even if all of the functionality has not been included. Customers are able to use and gain value from the software earlier than is possible with a waterfall process.

Reuse-oriented software engineering

In the majority of software projects, there is some software reuse. This often happens informally when people working on the project know of designs or code that are similar to what is required. They look for these, modify them as needed, and incorporate them into their system.

This informal reuse takes place irrespective of the development process that is used. However, in the 21st century, software development processes that focus on the reuse of existing software have become widely used. Reuse-oriented approaches rely on a large base of reusable software components and an integrating framework for the composition of these components. Sometimes, these components are systems in their own right (COTS or commercial off-the-shelf systems) that may provide specific functionality such as word processing or a spreadsheet.

A general process model for reuse-based development is shown in Figure 1.9. Although the initial requirements specification stage and the validation stage are comparable with

other software processes, the intermediate stages in a reuse oriented process are different. These stages are:

1. **Component analysis** Given the requirements specification, a search is made for components to implement that specification. Usually, there is no exact match and the components that may be used only provide some of the functionality required.
2. **Requirements modification** During this stage, the requirements are analyzed using information about the components that have been discovered. They are then modified to reflect the available components. Where modifications are impossible, the component analysis activity may be re-entered to search for alternative solutions.
3. **System design with reuse** During this phase, the framework of the system is designed or an existing framework is reused. The designers take into account the components that are reused and organize the framework to cater for this. Some new software may have to be designed if reusable components are not available.
4. **Development and integration** Software that cannot be externally procured is developed, and the components and COTS systems are integrated to create the new system. System integration, in this model, may be part of the development process rather than a separate activity.

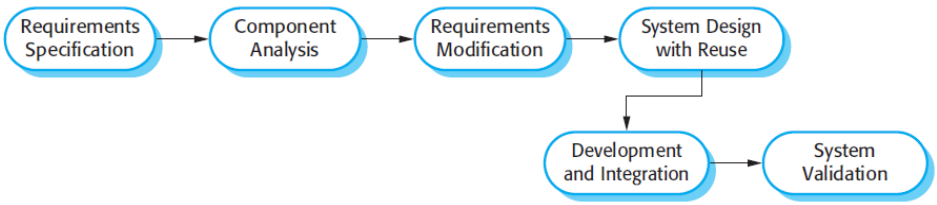


Fig. 1.9: Reuse-oriented Validation software engineering

There are three types of software component that may be used in a reuse-oriented process:

1. Web services that are developed according to service standards and which are available for remote invocation.
2. Collections of objects that are developed as a package to be integrated with a component framework such as .NET or J2EE.
3. Stand-alone software systems that are configured for use in a particular environment.

Advantages

Reuse-oriented software engineering has the obvious advantage of reducing the amount of software to be developed and so reducing cost and risks. It usually also leads to faster delivery of the software.

Disadvantages

The requirements compromises are inevitable and this may lead to a system that does not meet the real needs of users. Furthermore, some control over the system evolution is lost as

new versions of the reusable components are not under the control of the organization using them.

Spiral Model

The Spiral Model is a risk-driven software development process that combines elements of the Waterfall model and Iterative development. Proposed by Barry Boehm (1986), it focuses on continuous risk assessment and refinement through repeated cycles called spirals. It is especially useful for large, complex, and high-risk projects.

The **Spiral Model** is a risk-driven software development approach that combines iterative development with systematic risk analysis. Development proceeds through a series of loops called **spirals**, and each spiral represents one development cycle. During every cycle, the system is refined, risks are reduced, and user feedback is incorporated. Each spiral consists of **four major phases** described below.

1. Planning Phase

The planning phase begins each spiral cycle by defining the objectives and scope for that iteration. In this phase, the development team identifies the specific goals to be achieved, such as implementing a feature or improving system performance. Requirements relevant to the cycle are gathered and analyzed. Alternative approaches for achieving these objectives are explored, and estimates of time, cost, and resources are prepared. The output of this phase is a clear development plan and defined targets for the cycle.

2. Risk Analysis Phase

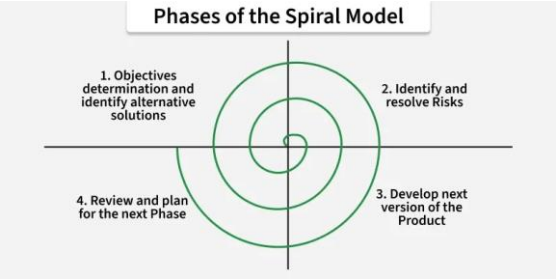
The risk analysis phase is the most critical part of the Spiral Model. Here, potential risks that could affect the project are identified and evaluated. These risks may be technical (new technology), financial (budget constraints), operational (performance issues), or schedule-related. To reduce uncertainty, prototypes or simulations may be developed, and feasibility studies may be conducted. Based on the analysis, strategies are prepared to minimize or eliminate risks. This phase ensures that major problems are addressed early before significant resources are invested.

3. Engineering Phase

During the engineering phase, the actual development work takes place. The system design is refined, and the planned features are implemented through coding. Developers build modules, integrate components, and conduct testing such as unit testing and integration testing to verify functionality. If prototypes were created earlier, they may be converted into working components. The outcome of this phase is a functional software increment that satisfies the objectives defined in the planning phase.

4. Evaluation Phase

The evaluation phase focuses on reviewing the results of the engineering work and obtaining feedback from stakeholders or customers. The developed increment is demonstrated and evaluated to determine whether it meets the intended objectives and user expectations. Users may suggest improvements or identify issues that need correction. Based on this feedback, a decision is made to proceed to the next spiral, revise the design, or, in rare cases, terminate the project. This phase ensures continuous improvement and alignment with user needs.



1.4.2 Software process activities

The four basic process activities of specification, development, validation, and evolution are organized differently in different development processes. In the waterfall model, they are organized in sequence, whereas in incremental development they are interleaved. How these activities are carried out depends on the type of software, people, and organizational structures involved. In extreme programming, for example, specifications are written on cards.

1.4.2.1 Software specification

Software specification or requirements engineering is the process of understanding and defining what services are required from the system and identifying the constraints on the system's operation and development. Requirements engineering is a particularly critical stage of the software process as errors at this stage inevitably lead to later problems in the system design and implementation.

The requirements engineering process (Figure 1.10) aims to produce an agreed requirements document that specifies a system satisfying stakeholder requirements. Requirements are usually presented at two levels of detail. End-users and customers need a high-level statement of the requirements; system developers need a more detailed system specification.

There are four main activities in the requirements engineering process:

1. **Feasibility study** an estimate is made of whether the identified user needs may be satisfied using current software and hardware technologies. The study considers whether the proposed system will be cost-effective from a business point of view and

if it can be developed within existing budgetary constraints. A feasibility study should be relatively cheap and quick. The result should inform the decision of whether or not to go ahead with a more detailed analysis.

2. **Requirements elicitation and analysis** This is the process of deriving the system requirements through observation of existing systems, discussions with potential users and procurers, task analysis, and so on. This may involve the development of one or more system models and prototypes. These help you understand the system to be specified.
3. **Requirements specification** Requirements specification is the activity of translating the information gathered during the analysis activity into a document that defines a set of requirements. Two types of requirements may be included in this document. User requirements are abstract statements of the system requirements for the customer and end-user of the system; system requirements are a more detailed description of the functionality to be provided.
4. **Requirements validation** This activity checks the requirements for realism, consistency, and completeness. During this process, errors in the requirements document are inevitably discovered. It must then be modified to correct these problems.

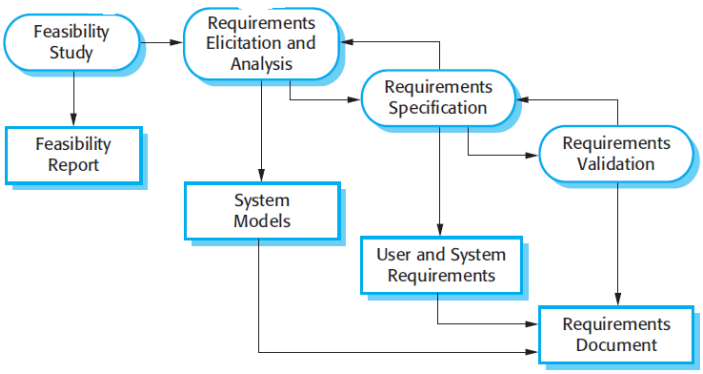


Fig. 1.10: The requirements engineering process

1.4.2.2 Software design and implementation

The **implementation stage** of software development is the process of converting a system specification into an executable system. It always involves processes of software design and programming but, if an incremental approach to development is used, may also involve refinement of the software specification.

A **software design** is a description of the structure of the software to be implemented, the data models and structures used by the system, the interfaces between system components and, sometimes, the algorithms used. Designers do not arrive at a finished design immediately but develop the design iteratively. They add formality and detail as they develop their design with constant backtracking to correct earlier designs.

Figure 1.11 is an abstract model of this process showing the inputs to the design process, process activities, and the documents produced as outputs from this process. The diagram suggests that the stages of the design process are sequential. In fact, design process activities are interleaved. Feedback from one stage to another and consequent design rework is inevitable in all design processes.

The **activities in the design process** vary, depending on the type of system being developed. For example, real-time systems require timing design but may not include a database so there is no database design involved. Figure 1.11 shows four activities that may be part of the design process for information systems:

1. **Architectural design**, where you identify the overall structure of the system, the principal components (sometimes called sub-systems or modules), their relationships, and how they are distributed.
2. **Interface design,** where you define the interfaces between system components. This interface specification must be unambiguous. With a precise interface, a component can be used without other components having to know how it is implemented. Once interface specifications are agreed, the components can be designed and developed concurrently.
3. **Component design**, where you take each system component and design how it will operate. This may be a simple statement of the expected functionality to be implemented. The design model may be used to automatically generate an implementation.
4. **Database design**, where you design the system data structures and how these are to be represented in a database. Again, the work here depends on whether an existing database is to be reused or a new database is to be created.

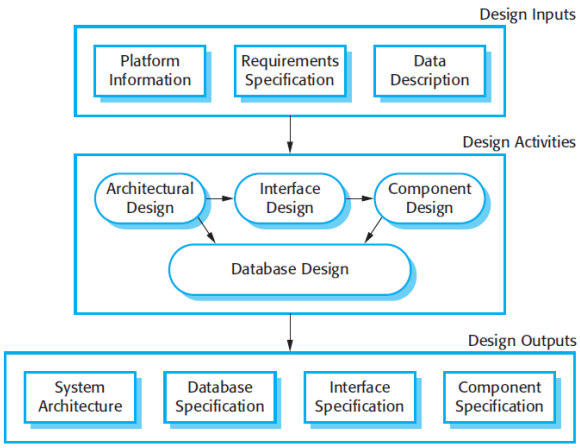


Fig. 1.11: A general model of the design process

1.4.2.3 Software validation

Software validation or, more generally, verification and validation (V&V) is intended to show that a system both conforms to its specification and that it meets the expectations of

the system customer. Program testing, where the system is executed using simulated test data, is the principal validation technique. Validation may also involve checking processes, such as inspections and reviews, at each stage of the software process from user requirements definition to program development. Because of the predominance of testing, the majority of validation costs are incurred during and after implementation.

Figure 1.12 shows a three-stage testing process in which system components are tested then the integrated system is tested and, finally, the system is tested with the customer's data. Ideally, component defects are discovered early in the process, and interface problems are found when the system is integrated.

The stages in the testing process are:

Development testing The components making up the system are tested by the people developing the system. Each component is tested independently, without other system components. Components may be simple entities such as functions or object classes, or may be coherent groupings of these entities. Test automation tools, such as JUnit, that can re-run component tests when new versions of the component are created, are commonly used.

System testing: System components are integrated to create a complete system. This process is concerned with finding errors that result from unanticipated interactions between components and component interface problems. It is also concerned with showing that the system meets its functional and non-functional requirements, and testing the emergent system properties.

Acceptance testing This is the final stage in the testing process before the system is accepted for operational use. The system is tested with data supplied by the system customer rather than with simulated test data. Acceptance testing may reveal errors and omissions in the system requirements definition, because the real data exercise the system in different ways from the test data. Acceptance testing may also reveal requirements problems where the system's facilities do not really meet the user's needs or the system performance is unacceptable.

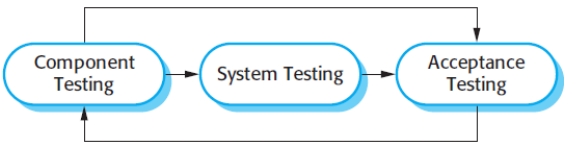


Fig. 1.12 Stages of testing

Figure 1.13 illustrates how test plans are the link between testing and development activities. This is sometimes called the V-model of development. Acceptance testing is sometimes called **'alpha testing'**. Custom systems are developed for a single client. The alpha testing process continues until the system developer and the client agree that the delivered system is an acceptable implementation of the requirements.

When a system is to be marketed as a software product, a testing process called '**beta testing**' is often used. Beta testing involves delivering a system to a number of potential customers who agree to use that system. They report problems to the system developers. This exposes the product to real use and detects errors that may not have been anticipated by the system builders. After this feedback, the system is modified and released either for further beta testing or for general sale.

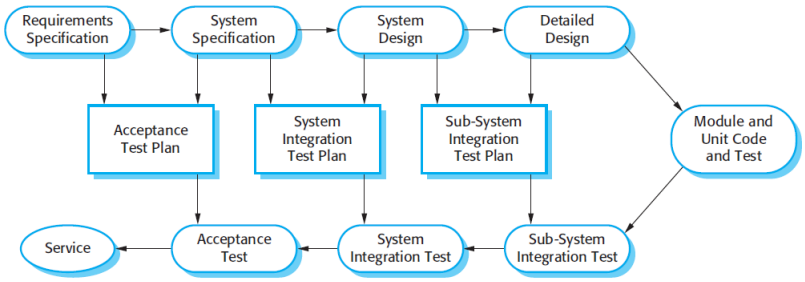


Figure 1.13: Testing phases in a plan-driven software process

1.4.2.4 Software evolution

The flexibility of software systems is one of the main reasons why more and more software is being incorporated in large, complex systems. Historically, there has always been a split between the process of software development and the process of software evolution (software maintenance). People think of software development as a creative activity in which a software system is developed from an initial concept through to a working system. However, they sometimes think of software maintenance as dull and uninteresting. Although the costs of maintenance are often several times the initial development costs, maintenance processes are sometimes considered to be less challenging than original software development.

This distinction between development and maintenance is increasingly irrelevant. Hardly any software systems are completely new systems and it makes much more sense to see development and maintenance as a continuum. Rather than two separate processes, it is more realistic to think of software engineering as an evolutionary process (Figure 1.14) where software is continually changed over its lifetime in response to changing requirements and customer needs.

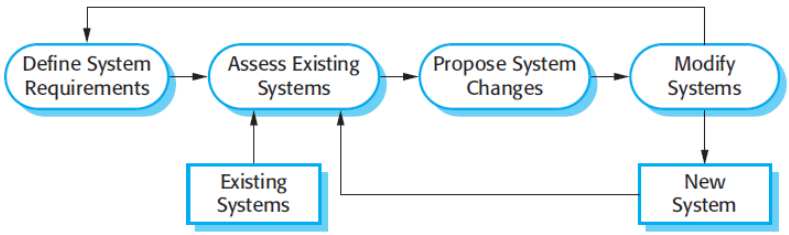


Figure 1.14 System evolution

1.5 Requirements Engineering

The **requirements** for a system are the descriptions of what the system should do, the services that it provides and the constraints on its operation. The process of finding out, analyzing, documenting and checking these services and constraints is called **requirements engineering (RE)**.

Requirement may range from a high-level abstract statement of a service or of a system constraint to a detailed mathematical functional specification.

- This is inevitable as requirements may serve a dual function
- May be the basis for a bid for a contract - therefore must be open to interpretation;
- May be the basis for the contract itself - therefore must be defined in detail;
- Both these statements may be called requirements.

Types of requirement

- ✧ User requirements
  - Statements in natural language plus diagrams of the services the system provides and its operational constraints. Written for customers.
- ✧ System requirements
  - A structured document setting out detailed descriptions of the system's functions, services and operational constraints. Defines what should be implemented so may be part of a contract between client and contractor.

Figure 1.15 illustrates the distinction between user and system requirements. This example from a mental health care patient management system (MHC-PMS) shows how a user requirement may be expanded into several system requirements.

User Requirement Definition

1. The MHC-PMS shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

System Requirements Specification

1.1 On the last working day of each month, a summary of the drugs prescribed, their cost, and the prescribing clinics shall be generated.

1.2 The system shall automatically generate the report for printing after 17.30 on the last working day of the month.

1.3 A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed, and the total cost of the prescribed drugs.

1.4 If drugs are available in different dose units (e.g., 10 mg, 20 mg) separate reports shall be created for each dose unit.

1.5 Access to all cost reports shall be restricted to authorized users listed on a management access control list.

Fig 1.15: User and system requirements

Figure 1.16 shows possible readers of the user and system requirements. The readers of the user requirements are not usually concerned with how the system will be implemented and may be managers who are not interested in the detailed facilities of the system. The readers of the system requirements need to know more precisely what the system will do because they are concerned with how it will support the business processes or because they are involved in the system implementation.

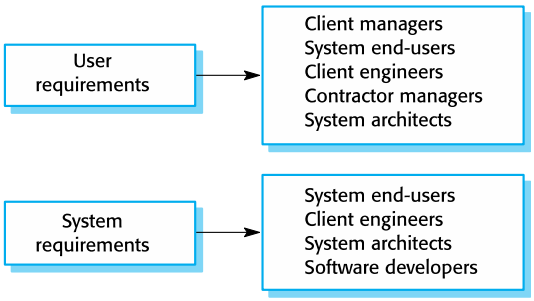


Figure 1.16: Readers of different types of requirements specification

1.5.1 Functional and non-functional requirements

1 **Functional requirements:** Statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations. In some cases, it may state what the system should not do.

2 **Non-functional requirements:** Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc. Often apply to the system as a whole rather than individual features or services.

3 **Domain requirements:** Constraints on the system from the domain of operation. Domain requirements reflect the environment in which the system operates so, when we talk about an application domain we mean environments such as train operation, medical records, e-commerce etc.

Functional requirements

- ✧ Describe functionality or system services.
- ✧ Depend on the type of software, expected users and the type of system where the software is used.
- ✧ Functional user requirements may be high-level statements of what the system should do.
- ✧ Functional system requirements should describe the system services in detail.

Mentcare system: functional requirements

- ✧ A user shall be able to search the appointments lists for all clinics.
- ✧ The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.

- Each staff member using the system shall be uniquely identified by his or her 8-digit employee number.

Requirements imprecision

- Problems arise when functional requirements are not precisely stated.
- Ambiguous requirements may be interpreted in different ways by developers and users.
- Consider the term 'search' in requirement
  - User intention – search for a patient name across all appointments in all clinics;
  - Developer interpretation – search for a patient name in an individual clinic. User chooses clinic then search.

Requirements completeness and consistency

- In principle, requirements should be both complete and consistent.
- Complete
  - They should include descriptions of all facilities required.
- Consistent
  - There should be no conflicts or contradictions in the descriptions of the system facilities.

Non-functional requirements

- These define system properties and constraints e.g. reliability, response time and storage requirements. Constraints are I/O device capability, system representations, etc.
- Process requirements may also be specified mandating a particular IDE, programming language or development method.
- Non-functional requirements may be more critical than functional requirements. If these are not met, the system may be useless.

Figure 1.17 is a **classification of non-functional requirements**. You can see from this diagram that the non-functional requirements may come from required characteristics of the software (product requirements), the organization developing the software (organizational requirements), or from external sources:

- **Product requirements:** These requirements specify or constrain the behavior of the software. Examples include performance requirements on how fast the system must

execute and how much memory it requires, reliability requirements that set out the acceptable failure rate, security requirements, and usability requirements.

- **Organizational requirements:** These requirements are broad system requirements derived from policies and procedures in the customer's and developer's organization. Examples include operational process requirements that define how the system will be used, development process requirements that specify the programming language, the development environment or process standards to be used, and environmental requirements that specify the operating environment of the system.
- **External requirements:** This broad heading covers all requirements that are derived from factors external to the system and its development process. These may include regulatory requirements that set out what must be done for the system to be approved for use by a regulator, such as a central bank; legislative requirements that must be followed to ensure that the system operates within the law; and ethical requirements that ensure that the system will be acceptable to its users and the general public.

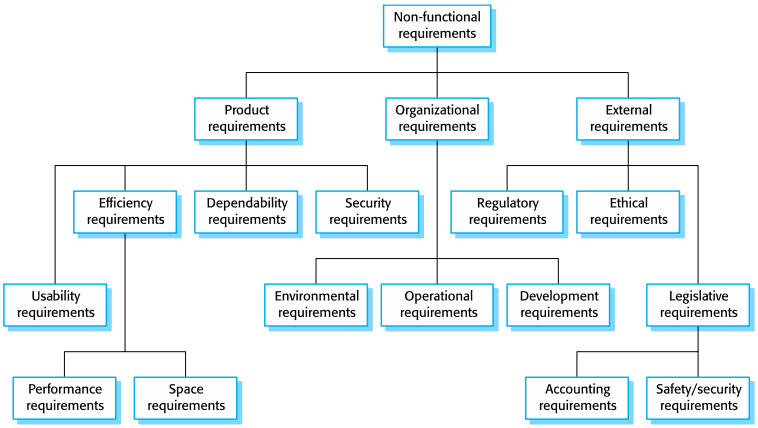


Fig. 1.17: Types of non-functional requirement

Figure 1.18 shows examples of product, organizational, and external requirements taken from the MHC-PMS. The product requirement is an availability requirement that defines when the system has to be available and the allowed down time each day. It says nothing about the functionality of MHC-PMS and clearly identifies a constraint that has to be considered by the system designers.

PRODUCT REQUIREMENT
The MHC-PMS shall be available to all clinics during normal working hours (Mon–Fri, 08.30–17.30). Downtime within normal working hours shall not exceed five seconds in any one day.

ORGANIZATIONAL REQUIREMENT
Users of the MHC-PMS system shall authenticate themselves using their health authority identity card.

EXTERNAL REQUIREMENT
The system shall implement patient privacy provisions as set out in HStan-03-2006-priv.

Fig.1.18: Examples of non-functional requirements in the MHC-PMS

Figure 1.19 shows metrics that you can use to specify non-functional system properties. You can measure these characteristics when the system is being tested to check whether or not the system has met its nonfunctional requirements.

Property	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Mbytes Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

Fig.1.19: Metrics for specifying non-functional requirements

1.5.2 The software requirements document

The software requirements document (sometimes called the software requirements specification or SRS) is an official statement of what the system developers should implement. It should include both the user requirements for a system and a detailed specification of the system requirements. Sometimes, the user and system requirements are integrated into a single description. In other cases, the user requirements are defined in an introduction to the system requirements specification. If there are a large number of requirements, the detailed system requirements may be presented in a separate document.

Requirements documents are essential when an outside contractor is developing the software system. However, agile development methods argue that requirements change so rapidly that a requirements document is out of date as soon as it is written, so the effort is largely wasted. For business systems where requirements are unstable, I think that this approach is a good one.

Figure 1.20 shows possible users of the document and how they use it. Figure 1.21 shows one possible organization for a requirements document that is based on an IEEE standard for requirements documents (IEEE, 1998). This standard is a generic standard that can be adapted to specific uses. In this case, I have extended the standard to include information about predicted system evolution. This information helps the maintainers of the system and allows designers to include support for future system features.

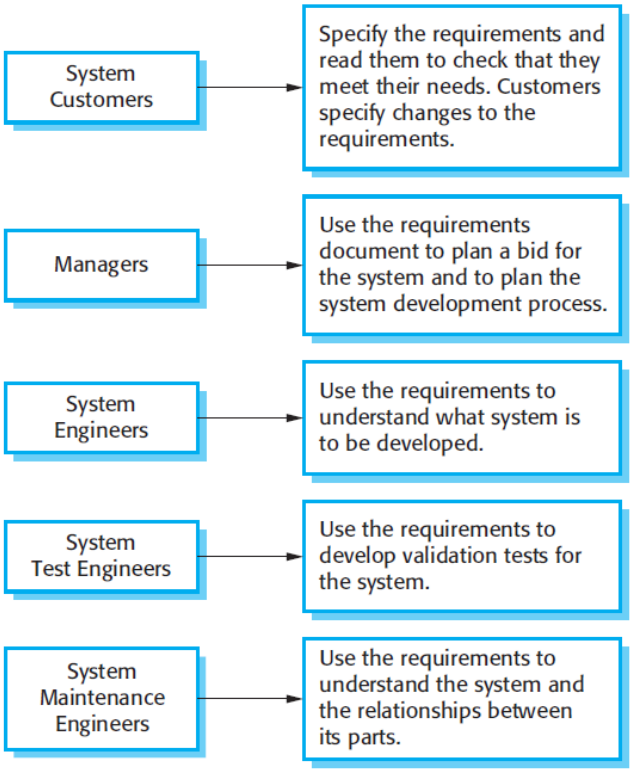


Fig.1.20: Users of a requirements document

Chapter	Description
Preface	This should define the expected readership of the document and describe its version history, including a rationale for the creation of a new version and a summary of the changes made in each version.
Introduction	This should describe the need for the system. It should briefly describe the system's functions and explain how it will work with other systems. It should also describe how the system fits into the overall business or strategic objectives of the organization commissioning the software.
Glossary	This should define the technical terms used in the document. You should not make assumptions about the experience or expertise of the reader.
User requirements definition	Here, you describe the services provided for the user. The non-functional system requirements should also be described in this section. This description may use natural language, diagrams, or other notations that are understandable to customers. Product and process standards that must be followed should be specified.
System architecture	This chapter should present a high-level overview of the anticipated system architecture, showing the distribution of functions across system modules. Architectural components that are reused should be highlighted.
System requirements specification	This should describe the functional and non-functional requirements in more detail. If necessary, further detail may also be added to the non-functional requirements. Interfaces to other systems may be defined.
System models	This might include graphical system models showing the relationships between the system components, the system, and its environment. Examples of possible models are object models, data-flow models, or semantic data models.
System evolution	This should describe the fundamental assumptions on which the system is based, and any anticipated changes due to hardware evolution, changing user needs, and so on. This section is useful for system designers as it may help them avoid design decisions that would constrain likely future changes to the system.
Appendices	These should provide detailed, specific information that is related to the application being developed; for example, hardware and database descriptions. Hardware requirements define the minimal and optimal configurations for the system. Database requirements define the logical organization of the data used by the system and the relationships between data.
Index	Several indexes to the document may be included. As well as a normal alphabetic index, there may be an index of diagrams, an index of functions, and so on.

Fig.1.21: The structure of a requirements document

1.5.3 Requirements specification

Requirements specification is the process of writing down the user and system requirements in a requirements document. Ideally, the user and system requirements should be clear, unambiguous, easy to understand, complete, and consistent. In practice, this

is difficult to achieve as stakeholders interpret the requirements in different ways and there are often inherent conflicts and inconsistencies in the requirements.

The **user requirements** for a system should describe the functional and nonfunctional requirements so that they are understandable by system users who don't have detailed technical knowledge. Ideally, they should specify only the external behavior of the system. The requirements document should not include details of the system architecture or design. Consequently, if you are writing user requirements, you should not use software jargon, structured notations, or formal notations. You should write user requirements in natural language, with simple tables, forms, and intuitive diagrams.

System requirements are expanded versions of the user requirements that are used by software engineers as the starting point for the system design. They add detail and explain how the user requirements should be provided by the system. They may be used as part of the contract for the implementation of the system and should therefore be a complete and detailed specification of the whole system.

Figure 1.22 summarizes the possible notations that could be used for writing system requirements. Graphical models are most useful when you need to show how a state changes or when you need to describe a sequence of actions.

Notation	Description
Natural language sentences	The requirements are written using numbered sentences in natural language. Each sentence should express one requirement.
Structured natural language	The requirements are written in natural language on a standard form or template. Each field provides information about an aspect of the requirement.
Design description languages	This approach uses a language like a programming language, but with more abstract features to specify the requirements by defining an operational model of the system. This approach is now rarely used although it can be useful for interface specifications.
Graphical notations	Graphical models, supplemented by text annotations, are used to define the functional requirements for the system; UML use case and sequence diagrams are commonly used.
Mathematical specifications	These notations are based on mathematical concepts such as finite-state machines or sets. Although these unambiguous specifications can reduce the ambiguity in a requirements document, most customers don't understand a formal specification. They cannot check that it represents what they want and are reluctant to accept it as a system contract.

Fig. 1.22: Ways of writing a system requirements specification

Natural language specification

Natural language has been used to write requirements for software since the beginning of software engineering. It is expressive, intuitive, and universal. It is also potentially vague, ambiguous, and its meaning depends on the background of the reader. As a result, there have been many proposals for alternative ways to write requirements. However, none of these have been widely adopted and natural language will continue to be the most widely used way of specifying system and software requirements. Figure 1.23 illustrates natural language specification for the embedded software for the automated insulin pump.

- The system shall measure the blood sugar and deliver insulin, if required, every 10 minutes.
- The system shall run a self-test routine every minute with the conditions to be tested and the associated actions.

Fig. 1.23: Example requirements for the insulin pump software system

Structured specifications

Structured natural language is a way of writing system requirements where the freedom of the requirements writer is limited and all requirements are written in a standard way. This approach maintains most of the expressiveness and understandability of natural language but ensures that some uniformity is imposed on the specification. Structured language notations use templates to specify system requirements. The specification may use programming language constructs to show alternatives and iteration, and may highlight key elements using shading or different fonts.

To use a structured approach to specifying system requirements, you define one or more standard templates for requirements and represent these templates as structured forms. The specification may be structured around the objects manipulated by the system, the functions performed by the system, or the events processed by the system. An example of a form-based specification, in this case, one that defines how to calculate the dose of insulin to be delivered when the blood sugar is within a safe band, is shown in Figure 1.24.

Insulin Pump/Control Software/SRS/3.3.2

Function	Compute insulin dose: Safe sugar level.
Description	Computes the dose of insulin to be delivered when the current measured sugar level is in the safe zone between 3 and 7 units.
Inputs	Current sugar reading (r2), the previous two readings (r0 and r1).
Source	Current sugar reading from sensor. Other readings from memory.
Outputs	CompDose—the dose in insulin to be delivered.
Destination	Main control loop.
Action	CompDose is zero if the sugar level is stable or falling or if the level is increasing but the rate of increase is decreasing. If the level is increasing and the rate of increase is increasing, then CompDose is computed by dividing the difference between the current sugar level and the previous level by 4 and rounding the result. If the result, is rounded to zero then CompDose is set to the minimum dose that can be delivered.
Requirements	Two previous readings so that the rate of change of sugar level can be computed.
Pre-condition	The insulin reservoir contains at least the maximum allowed single dose of insulin.
Post-condition	r0 is replaced by r1 then r1 is replaced by r2.
Side effects	None.

Fig. 1.24: A structured specification of a requirement for an insulin pump

Tables are particularly useful when there are several possible alternative situations and you need to describe the actions to be taken for each of these. The insulin pump bases its computations of the insulin requirement on the rate of change of blood sugar levels. The rates of change are computed using the current and previous readings. Figure 1.25 is a tabular description of how the rate of change of blood sugar is used to calculate the amount of insulin to be delivered.

Condition	Action
Sugar level falling (r2 < r1)	CompDose = 0
Sugar level stable (r2 = r1)	CompDose = 0
Sugar level increasing and rate of increase decreasing ((r2 − r1) < (r1 − r0))	CompDose = 0
Sugar level increasing and rate of increase stable or increasing ((r2 − r1) ≥ (r1 − r0))	CompDose = round ((r2 − r1)/4) If rounded result = 0 then CompDose = MinimumDose

Fig. 1.25: Tabular specification of computation for an insulin pump

1.5.4 Requirements engineering processes

The processes used for RE vary widely depending on the application domain, the people involved and the organization developing the requirements. However, there are a number of generic activities common to all processes:

- Requirements elicitation;
- Requirements analysis;
- Requirements validation;
- Requirements management.

In practice, requirements engineering is an iterative process in which the activities are interleaved. Figure 1.26 shows this interleaving. The activities are organized as an iterative process around a spiral, with the output being a system requirements document. This spiral model accommodates approaches to development where the requirements are developed to different levels of detail. The number of iterations around the spiral can vary so the spiral can be exited after some or all the user requirements have been elicited.

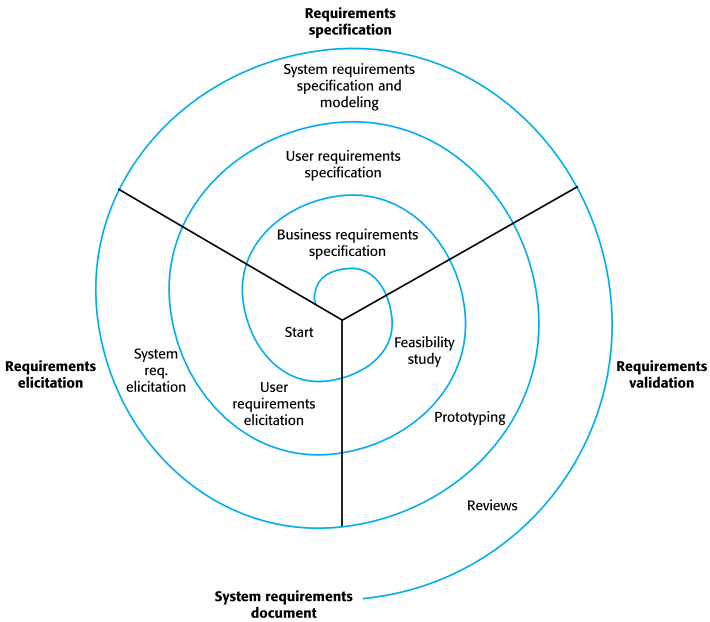


Fig. 1.26: A spiral view of the requirements engineering process

1.5.5 Requirements elicitation and analysis

After an initial feasibility study, the next stage of the requirement's engineering process is requirements elicitation and analysis. In this activity, software engineers work with customers and system end-users to find out about the application domain, what services the system should provide, the required performance of the system, hardware constraints, and so on.

A process model of the elicitation and analysis process is shown in Figure 1.27. Each organization will have its own version or instantiation of this general model depending on local factors such as the expertise of the staff, the type of system being developed, the standards used, etc. The process activities are:

1. **Requirements discovery:** This is the process of interacting with stakeholders of the system to discover their requirements. Domain requirements from stakeholders and documentation are also discovered during this activity.
2. **Requirements classification and organization:** This activity takes the unstructured collection of requirements, groups related requirements, and organizes them into coherent clusters. The most common way of grouping requirements is to use a model of the system architecture to identify sub-systems and to associate requirements with each sub-system.
3. **Requirements prioritization and negotiation:** Inevitably, when multiple stakeholders are involved, requirements will conflict. This activity is concerned with prioritizing requirements and finding and resolving requirements conflicts through negotiation. Usually, stakeholders must meet to resolve differences and agree on compromise requirements.
4. **Requirements specification:** The requirements are documented and input into the next round of the spiral.

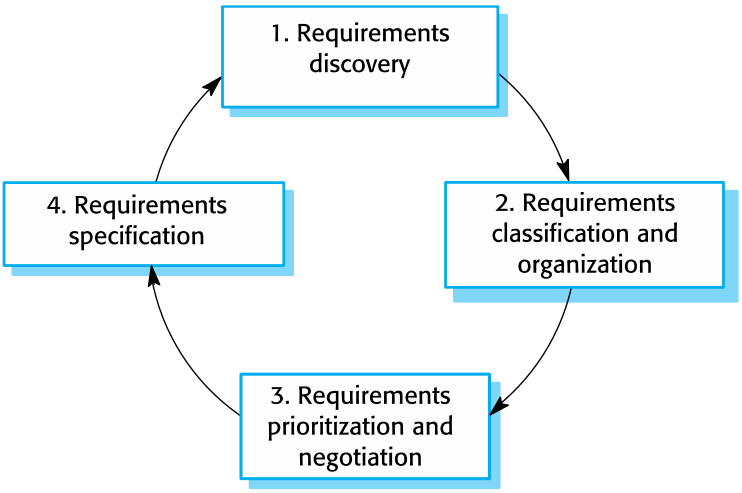


Fig. 1.27: The requirements elicitation and analysis process

Eliciting and understanding requirements from system stakeholders is a difficult process for several reasons:

1. Stakeholders often do not know what they want from a computer system except in the most general terms; they may find it difficult to articulate what they want the system to do; they may make unrealistic demands because they do not know what is and is not feasible.
2. Stakeholders in a system naturally express requirements in their own terms and with implicit knowledge of their own work. Requirements engineers, without experience in the customer's domain, may not understand these requirements.
3. Different stakeholders have different requirements and they may express these in different ways. Requirements engineers must discover all potential sources of requirements and discover commonalities and conflict.
4. Political factors may influence the requirements of a system. Managers may demand specific system requirements because these will allow them to increase their influence in the organization.
5. The economic and business environment in which the analysis takes place is dynamic. It inevitably changes during the analysis process. The importance of requirements may change. New requirements may emerge from new stakeholders who were not originally consulted.

1.5.5.1 Requirements discovery

Requirements discovery (sometime called requirements elicitation) is the process of gathering information about the required system and existing systems, and distilling the user and system requirements from this information. **Sources of information** during the requirements discovery phase include documentation, system stakeholders, and specifications of similar systems. You interact with stakeholders through interviews and observation and you may use scenarios and prototypes to help stakeholders understand what the system will be like.

Stakeholders range from end-users of a system through managers to external stakeholders such as regulators, who certify the acceptability of the system. For example, system stakeholders for the mental healthcare patient information system include:

1. Patients whose information is recorded in the system.
2. Doctors who are responsible for assessing and treating patients.
3. Nurses who coordinate the consultations with doctors and administer some treatments.
4. Medical receptionists who manage patients' appointments.
5. IT staff who are responsible for installing and maintaining the system.

6. A medical ethics manager who must ensure that the system meets current ethical guidelines for patient care.
7. Healthcare managers who obtain management information from the system.
8. Medical records staff who are responsible for ensuring that system information can be maintained and preserved, and that record keeping procedures have been properly implemented.

1.5.5.2 Interviewing

Formal or informal interviews with system stakeholders are part of most requirements engineering processes. In these interviews, the requirements engineering team puts questions to stakeholders about the system that they currently use and the system to be developed. Requirements are derived from the answers to these questions.

Interviews may be of two types:

1. **Closed interviews**, where the stakeholder answers a pre-defined set of questions.

2. **Open interviews**, in which there is no pre-defined agenda. The requirements engineering team explores a range of issues with system stakeholders and hence develop a better understanding of their needs.

It can be difficult to elicit domain knowledge through interviews for two reasons:

1 All application specialists use terminology and jargon that are specific to a domain. It is impossible for them to discuss domain requirements without using this terminology. They normally use terminology in a precise and subtle way that is easy for requirements engineers to misunderstand.
2 Some domain knowledge is so familiar to stakeholders that they either find it difficult to explain or they think it is so fundamental that it is not worth mentioning.

Effective interviewers have two **characteristics**:

They are open-minded, avoid pre-conceived ideas about the requirements, and are willing to listen to stakeholders. If the stakeholder comes up with surprising requirements, then they are willing to change their mind about the system.

They prompt the interviewee to get discussions going using a springboard question, a requirements proposal, or by working together on a prototype system.

1.5.5.3 Scenarios

Scenarios can be particularly useful for adding detail to an outline requirement description. They are descriptions of example interaction sessions. Each scenario usually covers one or a small number of possible interactions. Different forms of scenarios are developed and they provide different types of information at different levels of detail about the system.

A scenario starts with an outline of the interaction. During the elicitation process, details are added to this to create a complete description of that interaction. At its most general, a scenario may include:

1. A description of what the system and users expect when the scenario starts.
2. A description of the normal flow of events in the scenario.
3. A description of what can go wrong and how this is handled.
4. Information about other activities that might be going on at the same time.
5. A description of the system state when the scenario finishes.

1.5.5.4 Use cases

Use cases are a requirements discovery technique that were first introduced in the Objectory method. They have now become a fundamental feature of the unified modeling language. In their simplest form, a use case identifies the actors involved in an interaction and names the type of interaction. This is then supplemented by additional information describing the interaction with the system. The additional information may be a textual description or one or more graphical models such as UML sequence or state charts.

Use cases are documented using a high-level use case diagram. The set of use cases represents all the possible interactions that will be described in the system requirements. Actors in the process, who may be human or other systems, are represented as stick figures. Each class of interaction is represented as a named ellipse. Lines link the actors with the interaction. Optionally, arrowheads may be added to lines to show how the interaction is initiated. This is illustrated in Figure 1.28, which shows some of the use cases for the patient information system.

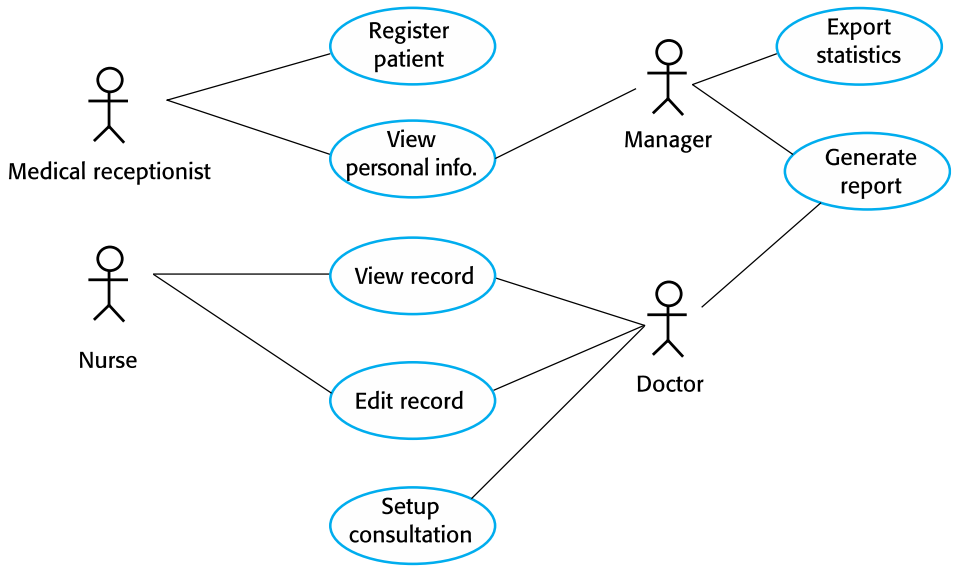


Fig. 1.28: Use cases Consultation for the MHC-PMS

1.5.6 Requirements validation

Requirements validation is the process of checking that requirements define the system that the customer really wants. It overlaps with analysis as it is concerned with finding problems with the requirements. Requirements validation is important because errors in a requirements document can lead to extensive rework costs when these problems are discovered during development or after the system is in service.

During the requirements validation process, different types of checks should be carried out on the requirements in the requirements document. These checks include:

1. **Validity checks:** A user may think that a system is needed to perform certain functions. However, further thought and analysis may identify additional or different functions that are required. Systems have diverse stakeholders with different needs and any set of requirements is inevitably a compromise across the stakeholder community.
2. **Consistency checks:** Requirements in the document should not conflict. That is, there should not be contradictory constraints or different descriptions of the same system function.
3. **Completeness checks:** The requirements document should include requirements that define all functions and the constraints intended by the system user.
4. **Realism checks:** Using knowledge of existing technology, the requirements should be checked to ensure that they can be implemented. These checks should also take account of the budget and schedule for the system development.
5. **Verifiability:** To reduce the potential for dispute between customer and contractor, system requirements should always be written so that they are verifiable. This means that you should be able to write a set of tests that can demonstrate that the delivered system meets each specified requirement.

There are several requirements validation techniques that can be used individually or in conjunction with one another:

1. **Requirements reviews:** The requirements are analyzed systematically by a team of reviewers who check for errors and inconsistencies.
2. **Prototyping**: In this approach to validation, an executable model of the system in question is demonstrated to end-users and customers. They can experiment with this model to see if it meets their real needs.
3. **Test-case generation:** Requirements should be testable. If the tests for the requirements are devised as part of the validation process, this often reveals requirements problems.

End of Module 1